



NORSK DATA

PRODUCT INFORMATION

Product name

COSMOS TCP/IP Gateway
for Ethernet

Product number

211185C

DESCRIPTION OF PRODUCT

To allow use of TCP/IP protocols by applications (FTP, TELNET) in a SINTRAN environment on a Ethernet local area network.

REASON(S) FOR NEW PRODUCT OR VERSION

- The Telnet server in COSMOS TCP/IP Gateway Version C07 is compatible with Telnet in Berkely Unix 4.3. It is not compatible with Telnet in Berkely Unix 4.2. Known dependencies are listed on page 3, other dependencies.
- It is possible to use the "get" and "pwd" FTP commands from a OPENLAN Ethernet/Cheapernet Adapter for Uniline 10/20/40 version A (Product number ND-110877) to FTP SERVER version C07.

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1 PREREQUISITES

The following things are required in order to use this product.

1.1 HARDWARE PREREQUISITES

The following hardware must be present for this product to be used:

CPU type (any of the following):

- 100 CX
- 110 CX
- 120 CX
- 500
- 5000

Floating format: *Any*

Other hardware:

Product name

Ethernet II controller

Product number

ND 110063

1.2 SOFTWARE PREREQUISITES

The following software must be installed on the system for this product to be used:

Operating system:

Name

SINTRAN III L06 PatchFile >= 2000B

Version/Revision

Other software:

Product name

BACKUP-SYSTEM version >=H

Product number

ND-210337

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1.3 OTHER PREREQUISITES

The following are necessary for this product to be used.

Minimum Permanent Mass Storage:

User	Space (pages)	Number of files
SYSTEM.....	200	5
TCP-IP.....	650	16

	Minimum	Maximum
Number of segments (ND-100)	3+4*NbOfControllers.	
Number of RT descriptions	3	

Space required on segment files: $120+256*NbOfControllers$ pages

Requirements regarding internal devices:

None

Requirements regarding batch processors:

None

2 DEPENDENCIES

In addition to the above listed hardware and/or software prerequisites, the following dependencies exist between this and other products. The hardware and software listed here may, for example, be necessary to use the product's full functionality.

2.1 HARDWARE DEPENDENCIES

There are dependency relationships between this product and the following hardware:

None

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2.2 SOFTWARE DEPENDENCIES

There are dependency relationships between this product and the following software:

Product nameProduct number

OWS Access Server for ND-500/5000

211325

Nature of dependency

The installation of the OWS Access Server for the ND-100 depends on this product to be installed..

2.3 OTHER DEPENDENCIES

In addition to the hardware and software dependency relationships listed above, you should be aware of the following dependencies:

The Telnet Server version C07 included in this product is according to the protocol in version 4.3 of Berkely Unix.

The Telnet Server version C07 is known to be compatible with the following products and versions:

211642	OpenLAN TIU TCP/IP boot SW II
211328	OpenLAN TIU TCP/IP boot SW
211643	Upgrade to TIU TCP/IP boot SW
211154	COSMOS TELNET/FTP clients (any version)

The Telnet Server version C07 is not compatible with:
Telnet client version 4.2 of Berkeley Unix.

3 DOCUMENTATION

Included (distributed free with product)TitleNumber

OpenLAN Network Supervisor Guide

30.107.03 EN

Optional (must be purchased)TitleNumber

COSMOS TELNET/FTP User Guide

60.284.01 EN

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4 INSTALLATION PROCEDURE

4.1 SELECTING INTERNET ADDRESS

- On request from the installation program you will have to assign an Internet address for each controller. If your net is isolated from other nets outside your company, you may select the address freely, i.e. using the default values given by the installation program. If your net connects to other nets, free selection of addresses may accidentally give the same address to two installations which will create severe problems. Selecting proper Internet addresses are further described in the OpenLAN Network Supervisor guide. However, the address for application of an official Internet address range is:

DDN Network Information Center
SRI International
Room EJ217
333 Ravenswood Avenue
Menlo Park, CA 94025

4.2 AUTOMATIC INSTALLATION

- Log in as user SYSTEM
- Insert the floppy 211185C-XX-01D in the floppy-drive
@ENTER-DIRECTORY,211185C,,
- Run the installation program:
@(211185:FLOPPY-USER)IN-TCP-IP

4.3 NON-AUTOMATIC INSTALLATION

When the installation program has been run:

- Update the AIP-HOSTS:SYMB file with the local internet address of the host name. See the documentation for more information.
- Update the mode file that is run after a cold start (usually called HENT-MODE:MODE) by including the following line:
@MODE (TCP-IP)TCP-IP-LO:MODE,,
If you had a previous version of this product, remove the line loading the product.

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- Update the mode file that is run after a warm-start (usually called LOAD-MODE:MODE) by including the following line:
`@MODE (TCP-IP)TCP-START:MODE,,`
 after XMSG and TADADM has started.
- Update the mode file that is run before stopping the machine (usually called STOP-MACHINE:MODE), by including the line:
`@MODE (TCP-IP)TCP-STOP:MODE,,`
- OWS Access Server If you have installed the OWS Access Server for ND100 (ND211297A07), include the line:
`@(OpenLAN)OWSAS-SERV START,,`
 in the LOAD-MODE:MODE file after the @START-SERVERS command, and
`@(OpenLAN)OWSAS-SERV STOP 0,,`
 in the @STOP-MACHINE:MODE file before @MODE (TCP-IP)TCP-STOP:MODE.
- Powerfail Handling
 The powerfail handler, POPWR, is an RT program that logs in on a Terminal Access Device on the local computer. POPWR will execute any mode file that does not stop TADs or XMSG. The hardcoded input and output file names are:
`(SYSTEM)PO-PWRFAIL:MODE` and `(SYSTEM)PO-PWRFAIL:OUTP`

To enable the program to be started at powerfail one must enter SINTRAN-SERVICE-PROGRAM and type:

```

DEFINE-USER-RESTART-PROGRAM  POPWR or
NEXT-USER-RESTART-PROGRAM POPWR   if the command
LIST-USER-RESTART-PROGRAM      gives any rt program
                                name as output.

```

Execution of the (TCP-IP)TCP-START:MODE file and any other mode files for devices that do not handle powerfail is then included in the PO-PWRFAIL:MODE file.

After the command @MODE (TCP-IP)TCP-IP-LO-C:MODE,, in the HENT-MODE file, the SINTRAN-SERVICE command:

```

@SINTRAN-SERVICE
@DEFINE-USER-RESTART-PROGRAM POPWR      or
@NEXT-USER-RESTART-PROGRAM POPWR      must be included.

```

Then the files PO-PWRFAIL:MODE and PO-PWRFAIL:OUTP must be created. User RT must have read and write access to these two files. Include in the PO-PWRFAIL:MODE file:

```

@MODE (TCP-IP)TCP-START-C:MODE,,

```

and the TCP/IP will be restarted after a powerfail.

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5 MODIFICATIONS

5.1 NEW FUNCTIONS/COMMANDS

- Responds to NetMap request from Network Control Server (NCS) and PCS Manager from 3Com.

5.2 CHANGED FUNCTIONS/COMMANDS

- FTP Server returns the name of the current directory inside '":
example: ftp> pwd 257 "PACK-ONE:MY-USER" is current directory. ftp>

This makes possible to use the "get" and "pwd" FTP commands from a OPENLAN Ethernet/Cheapernet Adapter for Uniline 10/20/40 version A (Product number ND-110877) to FTP SERVER version C07.

5.3 REMOVED FUNCTIONS/COMMANDS

None.

5.4 OTHER MODIFICATIONS

The PIOC-MONITOR has been included on diskette 2. The PIOC-MONITOR is meant for use by ND service personell only.

6 ERRORS CORRECTED

- TCP/IP can receive packets longer than 1118 bytes without stopping. The previous version of this software gave the error message: "TCP buffer already free" when receiving large packets i.e. UDP broadcasts.
- Telnet Server could display the error message: "Did not flush all the buffer in MTAD" when transferring a file from a workstation to the host. The error has been corrected.
- The FTP server can now transfer files up to 128 MB. Previous versions could not handle more than 16 MB.
- Earlier versions sometimes closed connections after 10 minutes without traffic. This was caused by an error in the no activity timer. This has been corrected in this version.

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211185C**7 ERRORS KNOWN BUT NOT CORRECTED**

- If a TCP/IP controller has been terminated due to an error, (for instance an HW error), a restart of TCP/IP may lead to a SINTRAN hang on level 14.
- Not all TCP/IP error messages are included in the UE-ERMSG file included on the diskettes, therefore, error messages may be printed with only error code, and not an error text.
- When communicating via TCP/IP using ND Socket Library (ND 211566), ICMP calls will lead to an XMSG message being reserved for each ICMP call done from an application. This may cause XMSG to run out of message buffers.